DTC P2105:00 [SKYACTIV-D 1.5]

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DTC P2105:00	**Intake air shutter valve actuator control system malfunction (forced engine shutdown)**
DETECTION CONDITION	• After switching the ignition off, the PCM detects that the engine speed has not decreased for a specified time. **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Engine oil deterioration • Fuel injector malfunction • Internal engine part malfunction (burning or seizure, abnormal combustion) • Crankshaft bearing malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**INSPECT ENGINE OIL LEVEL** • Inspect the engine oil level. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].) • Is the engine oil amount above X line on the oil level gauge?	Yes	Go to the next step.
		No	Go to Step 11.
4	**DETERMINE IF MALFUNCTION CAUSE IS BASE ENGINE OR OTHER** • Remove the water-cooled charge air cooler. (See WATER-COOLED CHARGE AIR COOLER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • When tilting the water-cooled charge air cooler, does engine oil drain from the water-cooled charge air cooler? **Note** • If the engine oil level has decreased only a small amount, go to Step “No” and replace the engine oil.	Yes	Go to the next step.
		No	Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) Perform the “ENGINE OIL DATA RESET”, then go to Step 11. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
5	**DETERMINE IF MALFUNCTION CAUSE IS ENGINE PART BURNING OR SEIZURE OR ABNORMAL COMBUSTION** • Remove the glow plug. (See GLOW PLUG REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Is the heat-generating area of the glow plug burnt out?	Yes	Go to the next step.
		No	Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) Perform the “ENGINE OIL DATA RESET”, then go to Step 11. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)

STEP	INSPECTION		ACTION
6	**VISUALLY INSPECT FOR FUEL INJECTOR AND CYLINDER HEAD** • Remove the fuel injector and cylinder head. (See FUEL INJECTOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].) (See CYLINDER HEAD GASKET REPLACEMENT [SKYACTIV-D 1.5].) • Visually inspect for the following. — There is nozzle deformation, damage to the opening of a fuel injector. — There are dents caused by piston impact, or adhesion of melt from glow plug, fuel injector, or piston on the cylinder head. • Is there any malfunction?	Yes	Go to the next step.
		No	Go to Step 9.
7	**VISUALLY INSPECT INSIDE OF CYLINDER** • Is there any vertical scratching (such as one that a fingernail can detect by scratching), or a dent on the cylinder liner caused by valve impact?	Yes	Replace or overhaul the engine as a partial engine unit. (See ENGINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Refer to the Mazda Workshop Manual.
		No	Go to the next step.
8	**INSPECT CRANKSHAFT BEARING FOR DAMAGE** • Remove the belts for the engine accessories. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Install a wrench to the crankshaft pulley lock bolt and turn it clockwise. (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].) • Does the crankshaft rotate smoothly?	Yes	If damage to the fuel injector is detected in Step 6: • Replace the fuel injector. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) If denting or adhesion of melted matter on the cylinder head is detected in Step 6: • Replace or overhaul the cylinder head. (See CYLINDER HEAD GASKET REPLACEMENT [SKYACTIV-D 1.5].) Add genuine engine oil, then go to Step 11. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
		No	Overhaul the engine, then go to Step 11.
9	**VISUALLY INSPECT INSIDE OF CYLINDER** • Is there any vertical scratching (such as one that a fingernail can detect by scratching), or a dent on the cylinder liner caused by valve impact?	Yes	Overhaul the engine, then go to Step 11.
		No	Go to the next step.
10	**INSPECT CRANKSHAFT BEARING FOR DAMAGE** • Remove the belts for the engine accessories. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Install a wrench to the crankshaft pulley lock bolt and turn it clockwise. (See FRONT OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].) • Does the crankshaft rotate smoothly?	Yes	Engine internal parts are normal. • Add genuine engine oil, then go to the next step. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
		No	Overhaul the engine, then go to the next step.

STEP	INSPECTION		ACTION
11	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the “FUEL INJECTOR INJECTION AMOUNT CORRECTION”. (See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].) • Start the engine and warm it up completely. **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Drive the vehicle under the FREEZE FRAME DATA/snapshot data condition. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
12	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.